Paper questions E.4: Fission [44 marks]

1. [Maximum mark: 1]
Three statements about a nuclear fission reactor are:

I. The heat exchanger transfers energy from the fuel rods to the moderator.
II. The control rods must be good absorbers of neutrons.
III. The moderator must slow neutrons down.

Which statements about the reactor are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme
C

2. [Maximum mark: 1]
Fuel rods in a nuclear fission reactor contain uranium isotopes U-235 and U-238. Which process taking place in the reactor contributes most significantly to the formation of radioactive waste products?

A. alpha decay of U-235

B. alpha decay of U-238

C. neutron-induced fission of U-235

D. neutron-induced transmutation of U-238 to plutonium-239 [1]

Markscheme
C

3. [Maximum mark: 1]
Three statements about the products of nuclear fission are:

I. some of them are chemically toxic
II. they have a wide range of half-lives
III. they release additional energy when removed from the reactor

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme
D

4. [Maximum mark: 1]
When removed from a nuclear reactor, used nuclear fuel rods are often stored for several years in a tank of liquid.

Three possible reasons for this are

I. to shield workers from radiation emitted from the fuel rods
II. to dissolve the unused fuel from the rods
III. to cool the rods

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I II and III [1]

Markscheme
B

5. [Maximum mark: 1]
Used fuel rods are stored in water after removal from a nuclear reactor.

The reason for this is that the water

A. absorbs the gamma radiation which is emitted by the used fuel

B. cools the used fuel which is still transferring energy

C. boils to steam which can then rotate a turbine

D. reacts chemically with the fuel which reduces the activity [1]

Markscheme
B

6. [Maximum mark: 8]

One possible fission reaction of uranium-235 $\left({}^{235}_{92}\mathrm{U}\right)$ is

$${}^{235}_{92}\mathrm{U} + n \rightarrow {}^{137}_{55}\mathrm{Cs} + {}^{95}_{37}\mathrm{Rb} + 4n$$

(a) Outline, with reference to the decay equation above, the role of chain reactions in the operation of a nuclear power station. [3]

Markscheme

The four neutrons released in the reaction may initiate further fissions ✓

«If sufficient U-235 is available,» the reaction is self-sustained ✓

Allowing for the continuous production of energy ✓

The number of neutrons available is controlled with control rods «to maintain the desired reaction rate» ✓

The following data are given:

Binding energy per nucleon of ${}^{235}_{92}\mathrm{U}$ = 7.591 MeV

Binding energy per nucleon of caesium-137 $\left({}^{137}_{55}\mathrm{Cs}\right)$ = 8.389 MeV

Binding energy per nucleon of rubidium-95 $\left({}^{95}_{37}\mathrm{Rb}\right)$ = 8.460 MeV

(b) Calculate, in MeV, the energy released in the reaction. [2]

Markscheme

$137 \times 8.389 + 95 \times 8.460 - 235 \times 7.591$ ✓

169 «MeV» ✓

(c) Two nuclides present in spent nuclear fuel $^{137}_{55}\mathrm{Cs}$ are and cerium-144 ($^{144}_{58}\mathrm{Ce}$). The initial activity of a sample of pure $^{144}_{58}\mathrm{Ce}$ is about 40 times greater than the activity of the same amount of pure $^{137}_{55}\mathrm{Cs}$.

Discuss which of the two nuclides is more likely to require long-term storage once removed from the reactor. [3]

Markscheme

«For the same number of nuclei,» the activity is inversely related to half-life ✓

Thus $^{137}_{55}\mathrm{Cs}$ has a longer half-life and will likely require longer storage ✓

Half-lives of their decay products need also be considered when planning storage ✓

7. [Maximum mark: 7]

(a) Compare and contrast spontaneous and neutron-induced nuclear fission. [2]

Markscheme

Spontaneous fission occurs with no external influence, neutron-induced fission requires an interaction with a neutron «of appropriate energy» ✓

Both result in the release of energy
OR
both have a large number of possible pairs of products ✓

(b) Every neutron-induced fission reaction of uranium-235 releases an energy of about 200 MeV. A nuclear power station transfers an energy of about 2.4 GJ per second.

Determine the mass of uranium-235 that undergoes fission in one day in this power station. [3]

Markscheme

Fissions per day $= \frac{2.4\times10^{9}\times60\times60\times24}{200\times10^{6}\times1.6\times10^{-19}}$ «$= 6.5\times10^{24}$» ✓

Mass of uranium $= \frac{6.5\times10^{24}}{6.05\times10^{23}} \times 0.235$ ✓

2.5 «kg» ✓

(c) State **two** properties of the products of nuclear fission due to which the spent nuclear fuel needs to be kept safe. [2]

Markscheme

Have relatively short half-lives / high activity ✓

Their decay products are «usually» also radioactive ✓

Volatile / chemically active ✓

Biologically active / easily absorbed by living matter ✓

8. [Maximum mark: 5]

(a) State **one** source of the radioactive waste products from nuclear fission reactions. [1]

Markscheme

Fission fragments from the fuel rods

OR activated materials in (e.g.) fuel rod casings

OR nuclei formed by neutron activation from U-235

OR stated products, e.g. Pu, U-236 etc. ✓

(b) Outline how this waste is treated after it has been removed from the fission reactor. [4]

Markscheme

Waste (fuel rod) is placed in cooling ponds for a number of years ✓

After most active products have decayed the uranium is separated to be recycled/reprocessed ✓

The remaining highly active waste is vitrified / made into a solid form ✓

And stored (deep) underground ✓

9. [Maximum mark: 13]

(a) Identify with ticks [✓] in the table, the forces that can act on electrons and the forces that can act on quarks.

	Weak nuclear	Strong nuclear
Electrons		
Quarks		

[2]

Markscheme

Weak nuclear: 2 ticks ✓
Strong nuclear: quarks only ✓

(b) The following data is available for atomic masses for the fusion reaction

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}:$$

$^{2}_{1}\mathrm{H}$	2.0141 u
$^{3}_{1}\mathrm{H}$	3.0160 u
$^{4}_{2}\mathrm{He}$	4.0026 u

(b.i) Show that the energy released is about 18 MeV. [2]

Markscheme

«☒» = 2.0141 + 3.0160 – (4.0026 + 1.008665) «= 0.0188 u»

OR

In MeV: 1876.13415 + 2809.404 – (3728.4219 + 939.5714475) ✓

= 0.0188 × 931.5 ***OR*** = 17.512 «MeV» ✓

Must see either clear substitutions or answer to at least 3 s.f. for ***MP2***.

(b.ii) Estimate the specific energy of hydrogen by finding the energy produced when 0.4 kg of $^{2}_{1}\mathrm{H}$ and 0.6 kg of $^{3}_{1}\mathrm{H}$ undergo fusion. [2]

Markscheme

ALTERNATIVE 1

0.40 kg of deuterium is «$\frac{400}{2}$ × 6.02 × 10^{23}» = 1.2 × 10^{26} nuclei
« 0.60 kg of tritium is the same number » ✓

So specific energy «$\frac{1.2\times10^{26}\times17.51\times10^{6}\times1.6\times10^{-19}}{0.4+0.6}$» = 3.4 × 10^{14} «J kg^{-1}»

ALTERNATIVE 2

«$17.51 \times 10^6 \times 1.6 \times 10^{-19}$ =» 2.8×10^{-12} «J»

AND

«$(2.0141 + 3.0160) \times 1.66 \times 10^{-27}$ =» 8.35×10^{-27} ✓

«$\frac{2.8 \times 10^{-12}}{8.35 \times 10^{-37}}$» = 3.4×10^{14} «J kg^{-1}»

Allow $\sim 2.1 \times 10^{27}$ MeV kg^{-1} *for* ***MP2***.

Allow ***ECF*** *from* ***MP1*** *for both* ***ALTs***.

(c) It is hoped that nuclear fusion can be used for commercial production of energy.

Outline

(c.i) **two** difficulties of energy production by nuclear fusion. [2]

Markscheme

Requires high temp/pressure ✓
Must overcome Coulomb/intermolecular repulsion ✓
Difficult to contain / control «at high temp/pressure» ✓
Difficult to produce excess energy/often energy input greater than output / ***OWTTE*** ✓
Difficult to capture energy from fusion reactions ✓
Difficult to maintain/sustain a constant reaction rate ✓

(c.ii) **one** advantage of energy production by nuclear fusion compared to nuclear fission. [1]

Markscheme

Plentiful fuel supplies ***OR*** larger specific energy ***OR*** larger energy density ***OR*** little or no «major radioactive» waste products ✓

Allow descriptions such as "more energy per unit mass" or "more energy per unit volume"

(d) Tritium ($^{3}_{1}\mathrm{H}$) is unstable and decays into an isotope of helium (He) by beta minus decay with a half-life of 12.3 years.

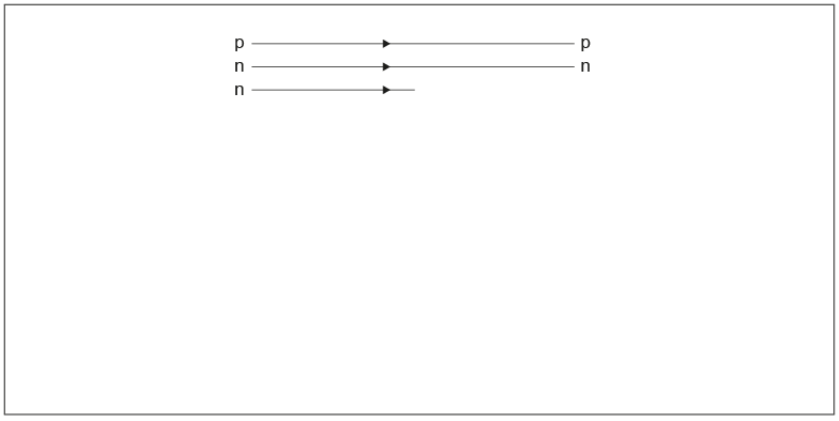
(d.i) State the nucleon number of the He isotope that $^{3}_{1}\mathrm{H}$ decays into. [1]

Markscheme

3 ✓

*Do **not** accept $^{3}_{2}He$ by itself.*

(d.ii) The following diagram is an incomplete Feynman diagram describing the beta minus decay of $^{3}_{1}\mathrm{H}$ into He. Complete the diagram and label all the missing particles.



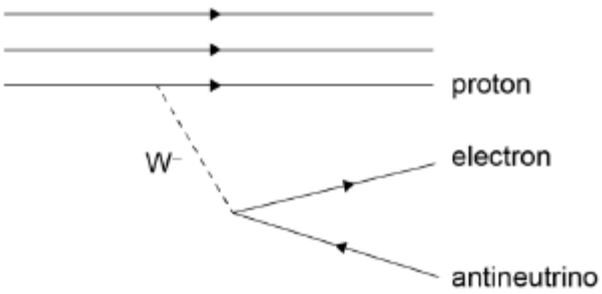
[3]

Markscheme

Proton shown ✓

W- shown ✓

Produces electron/e^- /$\boxtimes^-$ and antineutrino / $\overline{\upsilon}$ with proper arrow directions. ✓



Allow solid, dashed, or wavy line for W-particle.

Must see bar on antineutrino if symbol used.

10. [Maximum mark: 1]

A fusion reaction of one nucleus of hydrogen-2 and one nucleus of hydrogen-3 converts 0.019 u to energy. A fission reaction of one nucleus of uranium-235 converts a mass of 0.190 u to energy.

What is the ratio $\frac{\text{specific energy of this fusion of hydrogen}}{\text{specific energy of this fission of uranium}}$?

A. 0.1

B. 0.2

C. 5

D. 10 [1]

Markscheme
C

11. [Maximum mark: 1]

A neutron is absorbed by a nucleus of uranium-235 $\left({}^{235}_{92}\mathrm{U}\right)$. One possible outcome is the production of two nuclides, barium-144 $\left({}^{144}_{56}\mathrm{Ba}\right)$ and krypton-89 $\left({}^{89}_{36}\mathrm{Kr}\right)$.

How many neutrons are released in this reaction?

A. 0

B. 1

C. 2

D. 3 [1]

Markscheme
D

12. [Maximum mark: 1]

During the nuclear fission of nucleus X into nucleus Y and nucleus Z, energy is released. The binding energies per nucleon of X, Y and Z are

B_X, B_Y and B_Z respectively. What is true about the binding energy per nucleon of X, Y and Z?

A. $B_Y > B_X$ and $B_Z > B_X$

B. $B_X = B_Y$ and $B_X = B_Z$

C. $B_X > B_Y$ and $B_X > B_Z$

D. $B_X = B_Y + B_Z$

[1]

Markscheme

A

13. [Maximum mark: 1]

What are the principal roles of a moderator and of a control rod in a thermal nuclear reactor?

	Role of moderator	Role of control rod
A.	increases kinetic energy of neutrons	maintains a constant rate of reaction
B.	increases kinetic energy of neutrons	absorbs energy transferred in the reactor
C.	reduces kinetic energy of neutrons	maintains a constant rate of reaction
D.	reduces kinetic energy of neutrons	absorbs energy transferred in the reactor

[1]

Markscheme

C

14. [Maximum mark: 1]

What is the function of the moderator in a thermal nuclear fission reactor?

A. To decrease the kinetic energy of neutrons emitted from fission reactions

B. To increase the kinetic energy of neutrons emitted from fission reactions

C. To decrease the overall number of neutrons available for fission

D. To increase the overall number of neutrons available for fission [1]

Markscheme

A

15. [Maximum mark: 1]

A neutron collides head-on with a stationary atom in the moderator of a nuclear power station. The kinetic energy of the neutron changes as a result. There is also a change in the probability that this neutron can cause nuclear fission.

What are these changes?

	Change in kinetic energy of the neutron	Change in probability of causing nuclear fission
A.	increase	increase
B.	decrease	increase
C.	increase	decrease
D.	decrease	decrease

[1]

Markscheme

B

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